- **Set 1 ():** Orbital Boundaries, Openings/Foramina, Extraocular Muscles, and Orbital Neurovasculature.
- **Set 2 ():** Gross Anatomy of the Eyeball (Tunics, Refractive Media, Chambers, and Intraocular Vascular Supply).
- **Set 3 ():** Embryological Development of the Eye (Optic Cup Derivatives, Lens Placode, and Congenital Anomalies).

Set 1: Orbital Boundaries, Openings, and Contents (45 MCQs)

1. The bony orbit is a four-sided pyramidal cavity. The apex of this pyramid points postero-medially and opens directly into the middle cranial fossa through which structure?
a. Superior orbital fissure
b. Inferior orbital fissure
c. Optic canal
d. Foramen rotundum
e. Ethmoidal foramen
2. The roof (superior wall) of the orbit isolates the orbital contents from the anterior cranial fossa. This wall is formed primarily by the orbital plate of the frontal bone and the:
a. Greater wing of the sphenoid
b. Lesser wing of the sphenoid
c. Ethmoid bone
d. Zygomatic bone
e. Lacrimal bone
3. The floor (inferior wall) of the orbit is a thin plate of bone that is highly susceptible to blow-out fractures. It is formed mainly by the orbital surface of the:
a. Zygomatic bone
b. Maxilla
c. Palatine bone
d. Sphenoid bone
e. Ethmoid bone
4. The medial wall of the orbit is exceptionally thin and paper-like. The primary bone contributing to this wall, separating the orbit from the ethmoidal air sinuses, is the:
a. Lacrimal bone
b. Maxilla
c. Frontal bone
d. Lamina papyracea of the ethmoid
e. Greater wing of the sphenoid
5. The lateral wall of the orbit is the strongest of the orbital walls. Its anterior portion is formed by the:
a. Maxilla
b. Zygomatic bone
c. Frontal bone
d. Lesser wing of the sphenoid
e. Squamous temporal bone

6. The optic canal is a crucial opening located at the apex of the orbit within the lesser wing of the sphenoid. Which two structures pass through it?
a. Oculomotor nerve and ophthalmic vein
b. Optic nerve and ophthalmic artery
c. Ophthalmic nerve and ophthalmic vein
d. Trochlear nerve and lacrimal artery
e. Abducens nerve and central retinal vein
7. The superior orbital fissure lies between the greater and lesser wings of the sphenoid. Which nerve passes through this fissure *outside* the common tendinous ring (annulus of Zinn)?
a. Abducens nerve (CN VI)
b. Nasociliary nerve
c. Superior division of oculomotor nerve
d. Inferior division of oculomotor nerve
e. Frontal nerve (along with lacrimal and trochlear nerves)
8. Which cranial nerve passes directly through the central compartment of the superior orbital fissure, *inside* the common tendinous ring, to supply the lateral rectus muscle?
a. Trochlear nerve (CN IV)
b. Abducens nerve (CN VI)
c. Ophthalmic nerve (V1)
d. Oculomotor nerve (CN III)
e. Optic nerve (CN II)
9. The inferior orbital fissure is located in the floor of the orbit. It transmits which nerve branch heading toward the face?
a. Frontal nerve
b. Lacrimal nerve
c. Infraorbital nerve (from Maxillary nerve V2)
d. Nasociliary nerve
e. Mental nerve
10. The extraocular muscles control eyeball movements. Four of these muscles take their structural origin from a fibrous ring at the apex of the orbit called the:
a. Trochlea
b. Common tendinous ring (annulus of Zinn)
c. Periorbita sheet
d. Orbital septum
e. Lacrimal fascia
11. The lateral rectus muscle moves the eyeball out laterally (abduction). It is selectively innervated by which cranial nerve?
a. Oculomotor nerve (CN III)
b. Trochlear nerve (CN IV)
c. Trigeminal nerve (CN V)
d. Abducens nerve (CN VI)
e. Facial nerve (CN VII)
12. The superior oblique muscle passes through a fibrocartilaginous pulley attached to the frontal bone. It is innervated by which cranial nerve?
a. Oculomotor nerve (CN III)
b. Trochlear nerve (CN IV)

c. Abducens nerve (CN VI)
d. Nasociliary nerve
e. Facial nerve (CN VII)

13. Which extraocular muscle does *not* originate from the apex of the orbit, but instead takes its origin from the anteromedial floor of the orbit on the maxilla?

a. Inferior rectus
b. Superior oblique
c. Inferior oblique
d. Superior rectus
e. Medial rectus

14. The primary action of the superior rectus muscle when the eye is fully abducted is:

a. Depression
b. Intorsion
c. Extorsion
d. Elevation
e. Adduction

15. When testing the isolated action of the superior oblique muscle clinically, the patient is asked to look in which direction?

a. Upward and outward
b. Downward and outward
c. Downward and inward (adducted position)
d. Upward and inward
e. Purely laterally

16. The levator palpebrae superioris muscle elevates the upper eyelid. Its deep layer contains smooth muscle fibers (Müller's muscle) that receive innervation from the:

a. Oculomotor nerve
b. Facial nerve
c. Parasympathetic system
d. Sympathetic nervous system
e. Mandibular nerve

17. Loss of sympathetic innervation to the smooth muscle component of the levator palpebrae superioris results in partial dropping of the upper eyelid, a clinical sign known as:

a. Complete ptosis
b. Pseudoptosis
c. Lagophthalmos
d. Exophthalmos
e. Partial ptosis

18. The main arterial supply to the orbit and its contents is provided by the ophthalmic artery, which is a branch of the:

a. External carotid artery
b. Maxillary artery
c. Facial artery
d. Internal carotid artery
e. Middle meningeal artery

19. The first and most critical branch of the ophthalmic artery enters the optic nerve to run inside its core directly to the retina. This is the:

a. Ciliary artery
b. Lacrimal artery
c. Central retinal artery
d. Supraorbital artery
e. Ethmoidal artery

20. Occlusion or embolism of the central retinal artery leads to sudden, irreversible blindness. Because it does not anastomose with any other vessel, it is a classic example of an:

a. Anastomotic artery
b. Recurrent trunk
c. Collateral channel
d. Anatomical end artery
e. Sinusoidal capillary

21. Venous blood from the orbit is drained primarily by the superior and inferior ophthalmic veins. These vessels pass through the superior orbital fissure to empty into the:

a. Internal jugular vein
b. Pterygoid venous plexus
c. Cavernous sinus
d. Transverse sinus
e. Facial vein

22. Because the ophthalmic veins lack valves and communicate with both the facial vein and the cavernous sinus, infections from the face can track into the cranium, a hazard located in the:

a. Temporal fossa
b. Infratemporal fossa
c. Danger triangle of the face
d. Mandibular space
e. Sublingual compartment

23. The sensory nerve trunk of the orbit is the ophthalmic nerve (V1). Before entering the orbit, it divides into three major branches, which are the:

a. Infraorbital, zygomatic, palatine nerves
b. Lacrimal, frontal, and nasociliary nerves
c. Supraorbital, supratrochlear, infratrochlear nerves
d. Long ciliary, short ciliary, ethmoidal nerves
e. Buccal, mental, lingual nerves

24. The frontal nerve runs along the roof of the orbit and splits into two terminal branches that supply the skin of the forehead. These branches are the:

a. Supratrochlear and supraorbital nerves
b. Infratrochlear and lacrimal nerves
c. Zygomaticotemporal and buccal nerves
d. External nasal and internal nasal nerves
e. Anterior and posterior ethmoidal nerves

25. Which branch of the nasociliary nerve carries sensory fibers from the cornea and also contains postganglionic sympathetic fibers to the dilator pupillae muscle?

a. Short ciliary nerves
b. Infratrochlear nerve
c. Long ciliary nerves

d. Ethmoidal nerve
e. Lacrimal nerve

26. The ciliary ganglion is a small parasympathetic peripheral ganglion located at the apex of the orbit. Its preganglionic parasympathetic fibers travel via which nerve?

a. Optic nerve (CN II)
b. Oculomotor nerve (CN III - inferior division)
c. Nasociliary nerve (CN V1)
d. Abducens nerve (CN VI)
e. Sympathetic plexus

27. Postganglionic parasympathetic fibers leave the ciliary ganglion to travel to the eyeball to supply the sphincter pupillae and ciliary muscles via the:

a. Long ciliary nerves
b. Short ciliary nerves
c. Optic nerve sheath
d. Ophthalmic artery plexus
e. Lacrimal nerve

28. A patient is struck directly in the orbit by a baseball, causing a severe "blow-out" fracture. The orbital contents most commonly herniate downward into which adjacent cavity?

a. Frontal sinus
b. Ethmoidal air cells
c. Maxillary sinus
d. Nasal cavity proper
e. Infratemporal fossa

29. The lacrimal gland is located in the anterolateral roof of the orbit. Its secretomotor parasympathetic fibers are originally carried out of the brainstem by which cranial nerve?

a. Oculomotor nerve (CN III)
b. Trigeminal nerve (CN V)
c. Facial nerve (CN VII)
d. Glossopharyngeal nerve (CN IX)
e. Vagus nerve (CN X)

30. The postganglionic secretomotor parasympathetic fibers reach the lacrimal gland by hitchhiking along which final nerve branch?

a. Frontal nerve
b. Nasociliary nerve
c. Lacrimal nerve (after receiving fibers from the zygomaticotemporal nerve)
d. Infraorbital nerve
e. Supraorbital nerve

31. The fascial sheath that forms a socket for the eyeball, separating it from the orbital fat and allowing smooth global rotation, is known as the:

a. Periorbita
b. Orbital septum
c. Fascial sheath of eyeball (Tenon's capsule)
d. Suspensory ligament of Lockwood
e. Lacrimal fascia

32. Medial and lateral expansions of the extraocular muscle sheaths attach to the orbital walls to limit extreme movements of the eye. These are termed:

a. Suspensory bands
b. Check ligaments
c. Annular rings
d. Trochlear loops
e. Periorbital tracks

33. Injury to the oculomotor nerve (CN III) leads to complete paralysis of most extraocular muscles. Due to the unopposed action of the remaining healthy muscles, the eye deviates:

a. Downward and inward
b. Upward and outward
c. Downward and outward ("down-and-out")
d. Upward and inward
e. Purely medially

34. A patient presents with a complete lesion of the right abducens nerve. When asked to look straight ahead, the right eye deviates medially (internal strabismus) due to the unopposed action of the:

a. Superior rectus
b. Inferior rectus
c. Medial rectus
d. Superior oblique
e. Inferior oblique

35. The orbital septum is a thin fibrous membrane that hangs from the orbital margins. Its primary clinical importance is that it acts as a structural barrier preventing the spread of infection from:

a. The eyeball to the optic nerve
b. The eyelids (pre-septal) into the deep orbit (post-septal orbital cellulitis)
c. The maxillary sinus to the face
d. The lacrimal sac to the nasal cavity
e. The ethmoid sinus to the brain

36. The ophthalmic artery crosses either superior or inferior to which major structure as it travels from the apex toward the medial wall of the orbit?

a. Frontal nerve
b. Optic nerve
c. Abducens nerve
d. Lacrimal gland
e. Inferior oblique muscle

37. The branch of the ophthalmic nerve that passes through the anterior ethmoidal foramen to enter the cranial cavity before descending into the nasal cavity is the:

a. Infratrochlear nerve
b. Long ciliary nerve
c. Anterior ethmoidal nerve
d. Posterior ethmoidal nerve
e. External nasal nerve

38. The sensory root of the ciliary ganglion is derived from which nerve?

a. Oculomotor nerve
b. Frontal nerve
c. Nasociliary nerve
d. Lacrimal nerve

e. Trochlear nerve

39. The sympathetic fibers that enter the orbit to supply the dilator pupillae muscle arise from the superior cervical ganglion and travel along the wall of which artery?

a. External carotid artery
b. Ophthalmic artery
c. Internal carotid artery
d. Middle meningeal artery
e. Maxillary artery

40. A clinical condition characterized by abnormal protrusion or bulging of the eyeballs out of the orbital cavities, commonly seen in Graves' hyperthyroidism, is:

a. Enophthalmos
b. Ptosis
c. Blepharitis
d. Exophthalmos (Proptosis)
e. Chalazion

41. The condition of enophthalmos, where the eyeball is abnormally receded backward into the orbit, is a classic feature of which clinical neurological syndrome?

a. Bell's palsy
b. Horner's syndrome
c. Frey's syndrome
d. Lambert-Eaton syndrome
e. Wallenberg syndrome

42. The superior ophthalmic vein anastomoses anteriorly with which superficial vein of the face, providing a direct link for infection to spread?

a. Retromandibular vein
b. Maxillary vein
c. Facial vein (via the angular vein)
d. Superficial temporal vein
e. Occipital vein

43. The primary elevator of the upper eyelid, the levator palpebrae superioris, is located in which relative position within the orbit?

a. Along the medial wall
b. Deep along the floor
c. In the outermost lateral compartment
d. In the highest layer immediately beneath the roof
e. Wrapping around the posterior pole of the globe

44. The trochlear nerve (CN IV) enters the orbit through the superior orbital fissure outside the tendinous ring and travels medially to supply which muscle?

a. Superior rectus
b. Lateral rectus
c. Superior oblique
d. Inferior oblique
e. Levator palpebrae superioris

45. Testing the integrity of the afferent path of the corneal reflex involves checking which nerve branch that senses the cotton wisp touch?

a. Infraorbital nerve
b. Maxillary nerve
c. Ophthalmic nerve (via the nasociliary/long ciliary branches)

d. Mandibular nerve
e. Zygomatic nerve

Answer Key Table (

Q 1-8	Q 9-16	Q 17-24	Q 25-32	Q 33-40	Q 41-45

a	c	c	c	c	b
b	b	d	b	c	c
b	c	b	b	c	d
d	d	a	c	c	c
d	c	d	b	b	c
b	d	c	c	b	
a	e	a	b	c	
b	d	a	b	d	